Descriptive Statistics Primer

Objective

Calculate and interpret mean, median, mode, standard deviation, and percentiles for key numeric columns. The aim is to understand centre, spread, and possible skew in a dataset.

Dataset and method

This analysis uses the Iris teaching dataset: 150 observations, four numeric flower measurements (in centimetres), and a species label. The four numeric columns meet the task requirement of a 3–5 column summary. Standard deviation is the sample standard deviation (ddof=1); Q1 and Q3 are the 25th and 75th percentiles.

Summary statistics table

Numeric column	Count	Mean	Median	Mode	Std. dev.	Q1	Q3	Min	Max
Sepal Length (cm)	150	5.84	5.80	5.00	0.83	5.10	6.40	4.30	7.90
Sepal Width (cm)	150	3.06	3.00	3.00	0.44	2.80	3.30	2.00	4.40
Petal Length (cm)	150	3.76	4.35	1.40	1.77	1.60	5.10	1.00	6.90
Petal Width (cm)	150	1.20	1.30	0.20	0.76	0.30	1.80	0.10	2.50

What the numbers reveal: skew and spread

Sepal Length (cm): Mean and median are close, suggesting little obvious skew based on these two measures. Middle 50%: 5.10 to 6.40 cm (IQR 1.30 cm).

Sepal Width (cm): Mean and median are close, suggesting little obvious skew based on these two measures. Middle 50%: 2.80 to 3.30 cm (IQR 0.50 cm).

Petal Length (cm): Mean is below the median, which can indicate a left-skewed distribution or influence from smaller values. Middle 50%: 1.60 to 5.10 cm (IQR 3.50 cm).

Petal Width (cm): Mean is below the median, which can indicate a left-skewed distribution or influence from smaller values. Middle 50%: 0.30 to 1.80 cm (IQR 1.50 cm).

Overall narrative

The measurements have different centres and spreads, so they should not be treated as interchangeable. Mean and median provide a quick comparison for possible asymmetry, while standard deviation describes variability around the mean. The 25th and 75th percentiles show the middle 50% of values and the IQR summarises that spread robustly. Mean–median differences are clues rather than proof of skew; histograms or box plots should be checked before making a firm distribution claim.

Mini guide: when to use each statistic

Statistic	Meaning and use
Mean	Average; useful for overall level, but sensitive to outliers.
Median	Middle value; often better for skewed data or outliers.
Mode	Most frequently occurring value; useful for the most common value.
Standard deviation	Spread around the mean, expressed in the same units; interpret with the mean and context.
Percentiles / IQR	Q1 and Q3 bound the middle 50%; IQR = Q3 − Q1.

Tools and reproducibility

Tools covered: Excel, Python, and Pandas. The companion Excel workbook contains the full raw dataset, summary statistics, interpretation table, definitions, Python/Pandas examples, and Excel formula examples. Example formulas include AVERAGE, MEDIAN, MODE.SNGL, STDEV.S, and QUARTILE.INC.